

# Similar Matrices and Diagonalization

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# Learning Outcomes

At the end of this lecture, you should be able to:

- ☐ Know the properties of a diagonal matrix.
- ☐ Know what is meant by two matrices are similar and what does it implies.
- ☐ Know what is a diagonalizable matrix and determine if a matrix is diagonalizable.
- ☐ Find the diagonalization of a matrix.

# Diagonal matrix

Recall that a matrix  $D = [d_{ij}] \in M_n(\mathbb{R})$  is called a **diagonal matrix** if  $d_{ij} = 0$  for all  $i \neq j$ .

$$D = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & d_{nn} \end{bmatrix}$$

# Properties of diagonal matrix

Let  $D = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & d_{nn} \end{bmatrix}$  and  $W = \begin{bmatrix} w_{11} & 0 & \cdots & 0 \\ 0 & w_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & w_{nn} \end{bmatrix}$  be two diagonal matrices of order  $n$ .

1.  $DW = WD = \begin{bmatrix} w_{11}d_{11} & 0 & \cdots & 0 \\ 0 & w_{22}d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & w_{nn}d_{nn} \end{bmatrix}.$

2.  $\det D = d_{11}d_{22} \cdots d_{nn}$  and  $\det W = w_{11}w_{22} \cdots w_{nn}.$

# Properties of diagonal matrix

Let  $D = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & d_{nn} \end{bmatrix}$  and  $W = \begin{bmatrix} w_{11} & 0 & \cdots & 0 \\ 0 & w_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & w_{nn} \end{bmatrix}$  be two diagonal matrices of order  $n$ .

3. If all of the diagonal elements of  $D$  are nonzero, then  $D$  is invertible and

$$D^{-1} = \begin{bmatrix} \frac{1}{d_{11}} & 0 & \cdots & 0 \\ 0 & \frac{1}{d_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \frac{1}{d_{nn}} \end{bmatrix}.$$

4. The **eigenvalues** of  $D$  are its **diagonal elements**, that is  $d_{11}, \dots, d_{nn}$ .

$$|D| = \prod_{i=1}^n (d_{ii} - \lambda) = 0$$

$d_{ii} = \lambda_i$

# Similar matrices

Let  $A, B \in M_n(\mathbb{R})$ . A matrix  $A$  is **similar** to  $B$ , denoted by  $A \sim B$ , if there exists an invertible matrix  $P \in M_n(\mathbb{R})$  such that  $A = P^{-1}BP$ .

The similarity relation is an equivalence relation on  $M_n(\mathbb{R})$ .

$$\textcircled{1} R : A \sim A \text{ by } P = I.$$

$$\textcircled{2} S : A \sim B \rightarrow B \sim A \quad \begin{matrix} A = P^{-1}BP \\ PAP^{-1} = B \end{matrix}$$

$$\textcircled{3} T : \begin{matrix} A = P^{-1}AP \\ = R^{-1}CR \end{matrix}$$

# Properties of similar matrices

Let  $A, B \in M_n(\mathbb{R})$  be similar matrices. Then

$$A' = (P^{-1}BP)^{-1} = P^{-1}B^{-1}P$$

① If  $A$  and  $B$  are invertible, then  $A^{-1}$  and  $B^{-1}$  are similar.

②  $A^n$  and  $B^n$  are similar for any positive integer  $n$ .  $A^n = P^{-1}A^n P$

③  $A^T$  and  $B^T$  are similar.  $A^T = (P^{-1}BP)^T$ , let  $Q = P^T$   
 $= Q^T B Q$

④  $\text{adj}A$  and  $\text{adj}B$  are similar.  $\text{adj}A = |A| A'$

⑤  $\det A = \det B$ .  $|A| = |P^{-1}| |B| |P|$   
 $|A| = |B|$

⑥  $\rho(A) = \rho(B)$ .  
 $|A| \neq 0, \rho(A) = \lambda$

⑦  $A$  is invertible if and only if  $B$  is invertible.

⑧  $A$  and  $B$  have the same characteristics polynomial.

⑨  $A$  and  $B$  have the same eigenvalues.

$$I_n = P^{-1}P$$

$$\begin{aligned} |A - \lambda I_n| &= |P^{-1}BP - \lambda I_n| = |P^{-1}BP - \lambda P^{-1}P| \\ &= |P^{-1}(B - \lambda I_n)P| \\ &= |P^{-1}| |B - \lambda I_n| \cdot |P| \\ &= |B - \lambda I_n| \end{aligned}$$

# Diagonalization

Let  $A \in M_n(\mathbb{R})$ . Then  $A$  is said to be a **diagonalizable matrix**  $\Leftrightarrow$  there exists an invertible matrix  $P$  and diagonal matrix  $D$  such that  $P^{-1}AP = D$ .

If such a matrix  $P$  exists, we say that  **$P$  diagonalizes  $A$** .

Example:

Let  $A = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}$  and  $P = \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix}$ .

# Diagonalization

Example:

Show that  $A = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$  is not diagonalizable.

# Diagonalization

$$A \sim D \sim I$$

Let  $A \in M_n(\mathbb{R})$  be a diagonalizable matrix. Then

- 1  $A^n$  is diagonalizable for any positive integer  $n$ .
- 2  $A^T$  is diagonalizable.
- 3  $\text{adj}A$  is diagonalizable.

$$A = P^{-1}BP$$

$$\textcircled{1} A^n = P^{-1}B^nP \quad A^n \sim B^n$$

$$\textcircled{2} A^T = \underline{P^T} B^T (\underline{P^{-1}})^T$$



# Criteria for diagonalization

Let  $A \in M_n(\mathbb{R})$ . Then  $A$  is diagonalizable  $\Leftrightarrow A$  has  $n$  linearly independent eigenvectors.

$\rightarrow v \in \text{Eig}(\lambda) \text{ in } P$



$$P = [v_1 \ v_2 \ \dots \ v_n]$$

$\rightarrow |P| \neq 0$



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Let  $A \in M_n(\mathbb{C})$ . Then  $A$  has exactly  $n$  eigenvalues, counting all real and complex eigenvalues and counting multiplicities.  $F \neq A$

Let  $v_1, \dots, v_k$  be the eigenvectors of a matrix  $A$ . If each of them are corresponding to a different eigenvalue, then the eigenvectors are linearly independent.

$n$  column vector  $v_1, v_2, \dots, v_n$  are linearly independent  $\Leftrightarrow$   
 $A = [v_1, v_2, \dots, v_n]$  is invertible  $\Leftrightarrow \det A \neq 0$ .

If  $|A| = 0 \Leftrightarrow A$  is not dia.

$n \lambda \rightarrow |A| \neq 0$

# Diagonalization

To determine a matrix  $P$  that diagonalizes the matrix  $A$ :

1. Determine the characteristic equation of  $A$ ,  $p(\lambda) = \det(A - \lambda I_n) = 0$ .
2. Solve  $p(\lambda) = 0$  and determine all the eigenvalue  $\lambda_i$ .
3. For each  $\lambda_i$ , solve  $(A - \lambda_i I_n)v_i = 0$  to find an eigenvector  $v_i$  corresponding to  $\lambda_i$ .
4. Form the matrix  $P = [v_1 | v_2 | \dots | v_n]$
5. Determine  $|P|$ .

6. If  $|P| \neq 0$ , then  $P^{-1}AP =$  
$$\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix}.$$

# Diagonalization

Example:

Let  $A = \begin{bmatrix} -1 & 4 \\ 0 & 3 \end{bmatrix}$ . Find a matrix  $P$  that diagonalizes  $A$  and find the diagonal matrix  $D$  such that  $D = P^{-1}AP$

$$D = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$$

# Diagonalization

# Diagonalization

Example:

Let  $A = \begin{bmatrix} -1 & 0 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ . Find a matrix  $P$  that diagonalizes  $A$  and find the diagonal matrix  $D$  such that  $D = P^{-1}AP$

$$= \begin{bmatrix} -1 & & \\ & 1 & \\ & & -2 \end{bmatrix}$$

# Diagonalization

# Diagonalization

Example:

Let  $A = \begin{bmatrix} 5 & -4 & 4 \\ 12 & -11 & 12 \\ 4 & -4 & 5 \end{bmatrix}$ . Find a matrix  $P$  that diagonalizes  $A$  and find the diagonal matrix  $D$  such that  $D = P^{-1}AP$

# Diagonalization

# Calculating powers of matrices using diagonalization

Diagonalize  $A = \begin{bmatrix} 4 & -2 \\ -6 & 5 \end{bmatrix}$  and find a formula for  $A^k$ .

# Calculating powers of matrices using diagonalization

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Let  $A = \begin{bmatrix} 0 & 3 \\ 8 & 2 \end{bmatrix}$ .

- (a) Find the eigenvalues and eigenvectors of  $A$ .
- (b) Find the matrices  $P$  and  $D$  such that  $A = PDP^{-1}$ .
- (c) Evaluate  $A^n P$ .

# Calculating powers of matrices using diagonalization